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Film Cooling of a Gas Turbine Blade

The local film-cooling produced by a row of jets on a gas turbine blade is measured by a mass transfer technique. The density of the secondary fluid is from 0.75 to two times that of the mainflow and the range of the mass flux ratio is from 0.2 to three. The effect of blade-wall curvature on the film-cooling effectiveness is very significant. On the convex wall, a near tangential jet is pushed towards the wall by the static pressure force around the jet. For a small momentum flux ratio, this results in a higher effectiveness compared with that on a flat wall. At a large momentum flux ratio, however, the jet tends to move away from the curved wall because of the effect of inertia of the jet resulting in a smaller effectiveness on the convex wall. On the concave wall, the effects of curvature are the reverse of those described for the convex wall.

Introduction

Film cooling is used to protect a solid surface from a high temperature mainstream by releasing a coolant on the surface to be protected. This secondary fluid can be considered as a heat sink for the heat flow from the hot mainstream or as an insulating layer between the hot mainstream and the surface. A summary of film cooling on flat walls is given in [1].[2]

Film cooling has found wide application in gas turbines where performance generally increases with turbine inlet temperature. Although the strength of the alloys at high temperature has improved, a gas turbine provides a temperature and stress environment which is beyond the limit alloys can presently achieve without cooling. In the past, convection cooling was applied to blades and vanes. More recently film cooling combined with convection cooling and/or impingement cooling is used in blades and vanes.

For design purposes, it is often required to know the relation between the heat fluxes to a wall and the temperatures of the wall surface. With film cooling, the heat transfer coefficient is customarily defined by

$$q = h(T_w - T_{aw}) \tag{1}$$

The heat transfer coefficient and the adiabatic wall temperature have often been studied independently. The present study concerns the adiabatic wall temperature, T_{aw}, or the adiabatic film-cooling effectiveness, η_T, defined by

$$\eta_T = \frac{T_{aw} - T_\infty}{T_2 - T_\infty} \tag{2}$$

A mass-heat transfer analogy is useful in studying film-cooling effectiveness. Instead of injecting heated or cooled secondary fluid, a fluid at the mainstream temperature which contains a foreign gas is injected. A local impermeable-wall effectiveness, η, is defined by

$$\eta = \frac{C_{iw} - C_\infty}{C_2 - C_\infty} \tag{3}$$

If there is no foreign gas in the mainstream

$$\eta = \frac{C_{iw}}{C_2} \tag{4}$$

The analogy of the mass transfer process to the heat transfer process holds if the turbulent Lewis number and the molecular Lewis number are unity. If the flow is sufficiently turbulent, variations in the molecular Lewis number from unity may not play an important role [1]. Pedersen [2] checked the effect of molecular Schmidt number on the impermeable-wall effectiveness. There was no apparent effect of the molecular Schmidt number of the injection gas. Strictly speaking the analogy holds if the densities at arbitrary corresponding locations are the same in the two cases. For a turbulent flow, this requires that the molar heat capacities of the injected jet and the mainstream fluids are the same or that $T_2/T_\infty \cong 1$ [3]. When the analogy holds, the relation between the adiabatic film-cooling effectiveness and impermeable-wall effectiveness is given by

$$\eta_T = \frac{\eta}{\dfrac{c_{p\infty}}{c_{p2}} + \left(1 - \dfrac{c_{p\infty}}{c_{p2}}\right)\eta} \tag{5}$$

where $c_{p\infty}$ and c_{p2} are the specific heats of the mainstream and the secondary fluid in the heat transfer situation. This relation is verified [3] by using existing experimental results for two-dimensional film cooling. If $c_{p2} = c_{p\infty}$,

$$\eta_T = \eta \tag{6}$$

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[2] Numbers in brackets designate References at end of paper.

Contributed by the Gas Turbine Division of the AMERICAN SOCIETY OF MECHANICAL ENGINEERS and presented at the Tokyo Joint Gas Turbine Conference, Tokyo, Japan, May 23–27, 1977. Manuscript received at ASME Headquarters August 4, 1977.

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Most studies of film-cooling effectiveness have been done using plane walls. The effects of curvature have been investigated only recently.

Rastogi and Whitelaw [4] presented a procedure for predicting mean-flow properties of laminar and turbulent two-dimensional wall jets on curved walls. The paper briefly alluded to the effect of surface curvature on film cooling—indicating a higher effectiveness with a convex wall and a lower effectiveness with a concave wall. Folayan and Whitelaw [5] recently reported an experimental and computational investigation of the influence of curvature on the effectiveness of two-dimensional film cooling. Outside the range of parameters which results in separation, they indicate that an increase in convex curvature tends to improve effectiveness and an increase in concave curvature to decrease effectiveness.

Assuming that the static pressure distribution does not change with secondary fluid injection, Nicholas and LeMeur [6] predict from the momentum equations with a two-dimensional secondary fluid on a wall surface that if there is no longitudinal pressure gradiant, Archimedes type forces might either increase or decrease effectiveness depending upon the sign of the curvature of the wall and upon whether the momentum flux ratio is larger or smaller than unity. The effect of angle of injection is not included. For two-dimensional injection on a concave wall, they find good agreements with experimental results.

Hart [7] measured adiabatic film-cooling effectiveness downstream of injection through a single circular hole into a mainflow. The angle of injection was 35 deg on convex, flat, and concave walls. Up to a mass flux ratio of 0.4 to 0.5 the injection from the convex wall resulted in a higher centerline effectiveness than for the injection from the flat walls whereas on the concave wall the effectiveness was substantially lower. The curvature effect reversed at higher mass flux ratios.

Mayle, Kopper, Blair, and Bailey [8] studied curvature effects in two-dimensional film cooling. They found that the adiabatic film-cooling effectiveness was higher on a convex wall, but lower on a concave wall. This effect was attributed to the influence of curvature on Reynolds stress and the turbulent heat flux. The range of mass flux ratio was 0.5 to 0.9 at a density ratio of 0.9.

In the present paper measurements of the impermeable-wall effectiveness on convex, flat [2], and concave walls are compared. For the comparison the same ratio of boundary-layer thickness to the hole diameter would be ideal. However, though the effectiveness at the centerline of the hole is increased by decreasing the boundary layer displacement thickness [9, 10], the lateral average effectiveness is not greatly affected by the boundary displacement thickness (see [11] and [12]).

Analysis of Curvature Effects on Effectiveness

The following analysis will explain the curvature effect qualitatively and establish the important parameters.

Assumptions. The impermeable-wall effectiveness is given by equation (3) or (4). The wall gas concentration is affected by the path that the secondary (jet) flow follows as it penetrates into the primary flow. If the diffusion rates are the same for the same given conditions on walls with different curvatures, then one can expect that the jet whose height above the wall is smaller gives higher effectiveness if the velocities and the mass flow rates are the same.

The following assumptions are made in the analysis of the effect of curvature on film cooling:

(1) The radius of curvature of the wall, r_w, is constant along the downstream direction, X-direction.

(2) The amount of the secondary fluid is so small that the static pressure distributions are not disturbed by the secondary flow for a tangential jet.

(3) For a tangential jet, the momentum of the jet changes from $\rho_2 U_2^2$ at $X = 0$ to $\rho_\infty U_\infty^2$ for $X \to \infty$. For a jet injected at an angle to the surface, the density times the square of the X-component of the jet velocity changes monotonically from $\rho_2 U_2^2 \cos^2\alpha$ to $\rho_\infty U_\infty^2$ for 0 deg $\leq \alpha \leq$ 90 deg.

Jet Flow Over a Surface. Consider a jet flowing parallel to a wall. Over a flat wall the center of the jet trajectory tends to remain parallel to the wall neglecting entrainment effects near the surface of the wall. Over a curved wall the jet may move closer to or further from the wall. This tendency of the jet on a curved wall may be determined considering the balance of the forces exerted on a portion of the jet which

Nomenclature

C_{iw} = mass fraction of foreign gas at an impermeable wall
C_2 = mass fraction of foreign gas present in secondary fluid
C_∞ = mass fraction of foreign gas present in mainstream
c_{p2} = specific heat of secondary fluid
$c_{p\infty}$ = specific heat of mainstream fluid
D = diameter of injection hole
h = convective heat transfer coefficient
I = momentum flux ratio, $\rho_2 U_2^2/\rho_\infty U_\infty^2$
K = acceleration parameter, $\frac{\nu_\infty}{U_\infty^2}\frac{dU_\infty}{dX}$
L_c = blade chord length
L_s = blade pitch
M = blowing rate or mass flux ratio, $\rho_2 U_2/\rho_\infty U_\infty$
Ma_3 = Mach number at cascade exit
P = local static pressure
P_{tlc} = total pressure at center of cascade inlet
q = heat flow per unit time and area
Re_3 = chord Reynolds number, $U_3 L_c/\nu_3$
Re_∞ = mainstream Reynolds number based on hole diameter, $U_\infty D/\nu_\infty$
r = radius of curvature of mainstream. (The direction of radius is the same as that of r_w.)
r_j = radius of trajectory of center of injected fluid
r_w = radius of curvature of wall. (The positive direction of the radius is from the inside to the outside of the wall. $r_w > 0$ on convex wall, $r_w < 0$ on concave wall.)
r_{w0} = radius of curvature of wall at location of injection
T_{aw} = adiabatic wall temperature
T_w = wall temperature
T_2 = temperature of injected fluid
T_∞ = mainstream temperature or recovery temperature
t = spacing between the center lines of two adjacent injection holes
U = local mainstream velocity in flow direction
U_j = mean velocity of secondary fluid, downstream of an injection hole
U_1 = mean velocity of mainstream at cascade inlet
U_2 = mean velocity of secondary fluid at the exit of injection hole
U_3 = mean velocity of mainstream at cascade exit
U_∞ = mainstream velocity at location of injection
V_j = volume of a portion of secondary fluid, downstream of an injection hole
X = distance along wall downstream from the downstream edge of injection hole
X_n = normalized axial distance of cascade (cf. Fig. 3)
Z = lateral distance from centerline of injection hole
α = angle between the injection hole centerline and the direction of U_∞
δ^* = boundary layer displacement thickness at location of injection
η = local impermeable-wall effectiveness, equation (3)
$\bar{\eta}$ = lateral average of impermeable-wall effectiveness
η_T = local adiabatic film-cooling effectiveness
ν_3 = kinematic viscosity of mainflow at cascade exit
ν_∞ = kinematic viscosity of mainflow at injection
ρ_j = density of a portion of a jet, downstream of an injection hole
ρ_2 = density of secondary fluid at exit of injection hole
ρ_∞ = density of mainstream fluid
ϕ = positive value, $0 \leq \phi \leq 1$

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is assumed to be parallel to the wall at an arbitrary location to be considered.

Consider the flow over a curved wall with no jet present. The centrifugal force on a control volume would be balanced by the static pressure difference on the volume. Thus,

$$\frac{dP}{dr} = \rho_\infty \frac{U^2}{r} \tag{7}$$

where r is the radius of curvature of the stream. The direction of r and r_w are chosen to be positive from the inside of the wall to the outside. Then, r and r_w are positive for a convex wall and negative for a concave wall. If the total pressure is constant in the radial direction, U is inversely proportional to the absolute value of the radius.

If a jet is present and close to the wall as should be true for a film-cooling jet, the radius of curvature of the flow, r, is approximately r_w and the velocity (U) is U_∞. Then,

$$\frac{dP}{dr} = \rho_\infty \frac{U_\infty{}^2}{r_w} \tag{8}$$

Now consider a portion of a jet, downstream of an injection hole, which flows parallel to the wall (Fig. 1). The centrifugal force on the fluid in the control volume is in balance with the same force,

$$\frac{dP}{dr} = \rho_j \frac{U_j{}^2}{r_j} \tag{9}$$

In obtaining the trend of the jet trajectory we are ignoring the dynamic pressure of the mainstream on the jet which always tends to push the jet over into the direction of the mainstream.

From equations (8) and (9)

$$\rho_\infty \frac{U_\infty{}^2}{r_w} = \rho_j \frac{U_j{}^2}{r_j}$$

or

$$\frac{r_j}{r_w} = \frac{\rho_j U_j{}^2}{\rho_\infty U_\infty{}^2} \tag{10}$$

The value of $\rho_j U_j{}^2$ should be between the value of the entering jet $\rho_2 U_2{}^2$ and the value $\rho_\infty U_\infty{}^2$. Thus, with

$$\phi = \frac{\rho_j U_j{}^2 - \rho_\infty U_\infty{}^2}{\rho_2 U_2{}^2 - \rho_\infty U_\infty{}^2}$$

$$\frac{\rho_j U_j{}^2}{\rho_\infty U_\infty{}^2} = 1 + \phi(I - 1) \tag{11}$$

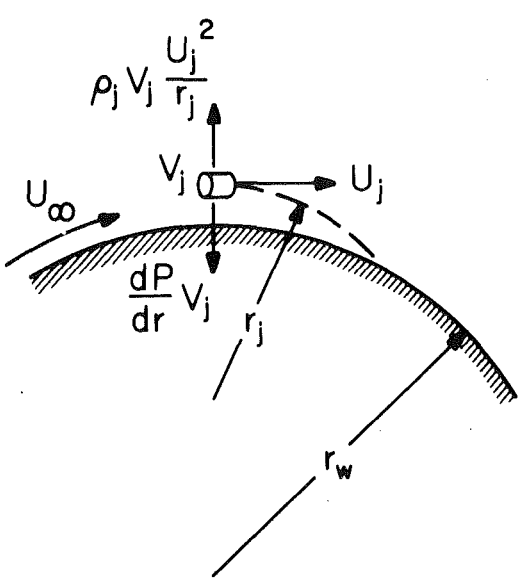

Fig. 1 The center of the trajectory of a jet flowing over a convex wall

where

$$0 \le \phi \le 1$$

Putting this into equation (10)

$$\frac{r_j}{r_w} = 1 + \phi(I - 1) \tag{12}$$

The relative size of r_j is dependent on the momentum flux ratio. Thus

$$|r_j| > |r_w| \quad \text{for } I > 1$$

$$|r_j| = |r_w| \quad \text{for } I = 1$$

$$|r_j| < |r_w| \quad \text{for } I < 1$$

Considering the direction of the static pressure force over a surface, the sign of the curvature of the jet and the wall should be the same. If the absolute value of the radius of the jet is smaller than that of a curved wall, the jet comes closer to a convex wall, but moves away from a concave wall. Then, the film-cooling effectiveness on a convex wall for given conditions would be better than that on the flat or concave wall under the same conditions if $I < 1$ and worse if $I > 1$. On a concave wall, the film-cooling effectiveness for given conditions is better than that on a flat or convex wall if $I > 1$ and worse if $I < 1$.

If a film-cooling jet enters the mainflow at some angle, α, (0 deg $\le \alpha \le$ 90 deg) the motion of that jet over a curved wall can be used to determine the relative film-cooling performance. For the height of a jet above a curved surface to be the same at any location as the height of a jet leaving a flat surface, the force in the radial direction due to the static pressure on the jet must balance the force to accelerate the jet in the radial direction. Then results analogous to those found above for a jet entering tangentially are found if $\rho_2 U_2{}^2 \cos^2\alpha$ is used instead of $\rho_2 U_2{}^2$. Again we are ignoring the effect of the dynamic pressure of the mainstream. This should not affect the trend of the change in the direction and thus the trend in cooling effectiveness on walls of different curvature. When $I \cos^2\alpha$ is less than unity the resulting jet would tend to turn more towards the surface on a convex wall than on a flat or a concave wall.

The film-cooling effectiveness on a convex wall is better than on a flat or a concave wall if $I \cos^2\alpha < 1$ and worse if $I \cos^2\alpha > 1$ for the same conditions. The film-cooling effectiveness on a concave wall is better than that on a flat wall if $I \cos^2\alpha > 1$ and worse if $I \cos^2\alpha < 1$ for 0 deg $< \alpha <$ 90 deg.

If a jet were injected upstream ($\alpha >$ 90 deg), the injected jet would tend to be turned downstream a short distance from the location of injection. Then the effect of inertia of the jet on attachment or separation from the curved surface would be small; instead, the jet is mainly influenced by the static pressure force on it. The film-cooling effectiveness on a convex wall should be then better than on a flat or concave wall at all blowing rates.

Experimental Apparatus

The present experimental study has been carried out using a low velocity wind tunnel at room temperature as shown in Fig. 2. A cascade of six blades—four solid and two hollow blades—is used. One of the hollow blades has a row of holes (α = 35 deg) on the suction side and the other has a row of holes α = 35 deg on the pressure side. The surfaces with holes face each other. The spacing between the centerlines of two adjacent injection holes is $3D$ (D = 0.238 cm). The ratio of blade span to chord length is 3.55. The rather long span is designed to avoid secondary flow and to allow use of only the central part of the span where the inlet velocity distribution is expected to be uniform. There are threaded holes in the top wall 0.76 times the chord length upstream of the leading edges of the blades to insert pitot tubes and a thermocouple. The inlet velocities, angles and temperature are measured at this section.

The shape of the blade used in the present study was supplied by the Aircraft Gas Turbine Division of the General Electric Co. The blade chord length, L_c, and the pitch, L_s, are 16.91 cm and 13.11 cm respectively. This compares to the chord length of 4.0 cm for the actual

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blade. The secondary fluid is supplied to the inside of the blade through a brass tube which has a row of holes. Taps for obtaining gas samples are located near the center of the span. Samples of gas can be drawn from all the taps in a blade at once. To assure spanwise uniformity of the jets, the brass tube carrying the secondary gas in the blade is rotated and fixed at the position giving the most uniform flow. Satisfactory uniformity was obtained. In most of the experiments only 23 of the 79 holes are used. The rest of the holes are covered by thin tapes.

Fig. 3 shows the normalized mainstream velocity profiles on the blade surfaces. The velocities are obtained from the inlet total pressure at the center of the inlet section and the static pressure at the taps in the walls assuming that the flow is incompressible and potential flow. This is actually $\sqrt{2(P_{tlc} - P)/\rho_\infty}/U_3$. The experimental results agree well with the calculated results which were supplied by the General Electric Co. The measured inlet and outlet flow angles also agree within 0.2 to 0.3 deg with the calculated flow angles.

The secondary gas is produced with the desired density from a mixture of helium or refrigerant-12 and air. The sampled gases are analyzed by a gas chromatograph. Details of the secondary and sampling systems and the analysis of sampling gases are given in [2, 3].

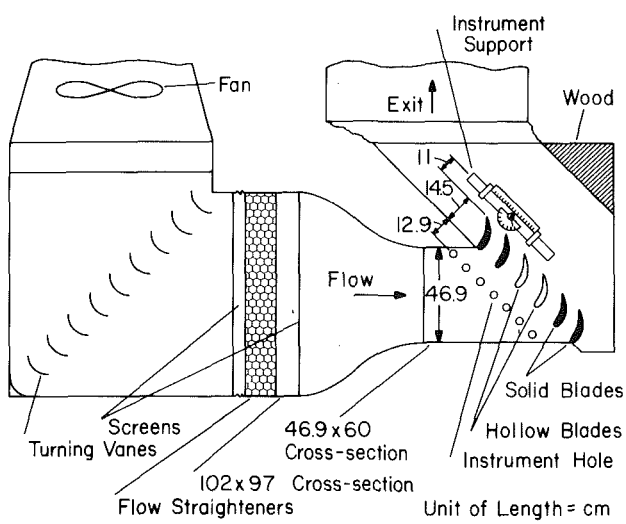

Fig. 2 Schematic of wind tunnel and cascade

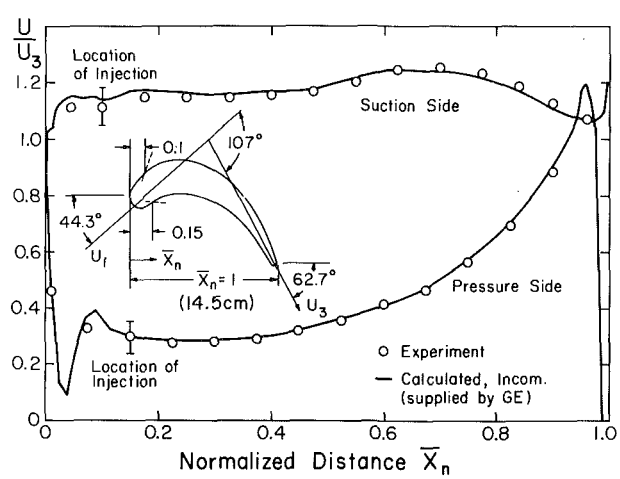

Fig. 3 Normalized mainstream velocity profile on blade, $Ma_3 = 0.065$, $Re_3 = 2.32 \times 10^5$, $\alpha = 35$ deg

At the location of injection on the suction side, the calculated boundary layer displacement thickness, δ^*, (using the approximate method due to von Kármán and Pohlhausen [13]) is 0.22 mm and δ^*/D is approximately 0.09. With this δ^*, $U_\infty\delta^*/\nu_\infty = 300$. Since the critical Reynolds number for $K = 0$ is about 660 [13], the boundary layer before the injection holes is expected to be laminar or in transition. On the pressure side, there is an adverse pressure gradient near the nose of the blade. The boundary layer at the injection hole on this surface is expected to be turbulent which is confirmed by the experiments of the aerodynamic loss of the flow in the cascade [3].

Experimental Results and Discussions

If the boundary layer is not turbulent, an additional requirement for fulfilling the mass-heat transfer analogy is that the value of the molecular Schmidt number is equal to the Prandtl number. However, even though these are different, in film-cooling the analogy may still be fulfilled because of the strong turbulent mixing of the secondary fluid with the mainstream. Also, since the gradient of the mass concentration (temperature) perpendicular to the wall should be zero on the impermeable (adiabatic) wall, the effects of the laminar region may be very small. The effects of Schmidt number on impermeable-wall effectiveness are checked [3] by using air with a tracer of helium, whose molecular weight is 4 and Schmidt number is 0.2 in the air, and refrigerant-12, whose molecular weight is 120.9 and Schmidt number is 1.7 in air [2]. The effectiveness with the different tracers was similar for similar experimental conditions on both pressure and suction surfaces—indicating that the mass-heat transfer analogy holds in the present experiments.

The effect of the mainstream Reynolds number on the lateral average impermeable-wall effectiveness on the suction (convex) side was checked by reducing the mainstream speed to about half of that normally used in the experiments. The average effectiveness depended very little on the mainstream Reynolds number.

Examples of the results of local impermeable-wall effectiveness measurements for the convex (suction) and concave (pressure) surfaces are shown in Figs. 4 and 5.

The lateral average effectiveness for the density ratio of 0.95 as a function of X/D is shown in Fig. 6 for suction side injection and in Fig. 7 for pressure side injection. On the suction side the largest effectiveness is obtained at $M = 0.5$ to 0.7. With further increase of M, the effectiveness decreases because of penetration of jets through the boundary layer. For M from 1.5 to 2.0 the effectiveness increases because of increase of the secondary mass flow even though the jets may penetrate through the boundary layer. On the pressure side the effectiveness at small M (or I) is very small. This is because the static pressure force acts on the jets in the direction to move the jets away from the surface. With high M or I for the near tangential jets the jets tend to strike the concave surface downstream resulting in relatively high effectiveness.

In Figs. 8 and 9, comparisons of the results for convex, flat, and concave walls are shown at $\rho_2/\rho_\infty = 0.95$ and $\rho_2/\rho_\infty = 2.0$ respectively. There are substantial differences for the different surfaces. The importance of $I \cos^2\alpha$ as a parameter to correlate the curvature effect on the effectiveness was shown in the analysis. For $\rho_2/\rho_\infty = 2.0$ with $I \cos^2\alpha = 1.3$ (see Fig. 9), the average effectiveness is smallest on the concave wall near the injection hole. At that condition, the kidney-shaped jet may just be forming. Thus, the effectiveness is very sensitive to experimental conditions. Further downstream the concave surface gives higher effectiveness.

For a better understanding of the curvature effects, the effectiveness is shown as a function of $I \cos^2\alpha$ at $X/D = 10$ and 40 for $\rho_2/\rho_\infty = 0.95$ in Fig. 10. The lines for convex, flat, and concave surfaces cross at $I \cos^2\alpha \cong 1$.

Fig. 11 shows comparisons of the average wall effectiveness at a mass flux ratio of 0.5 for different density ratios on flat, convex, and concave surfaces. At the lowest density ratio the jet has a larger momentum flux and thus tends to move further away from the surface resulting in lower effectiveness for the same mass flow (M). This can be observed for the results for the flat and concave walls. However, on the convex wall this trend is observed only near the injection holes.

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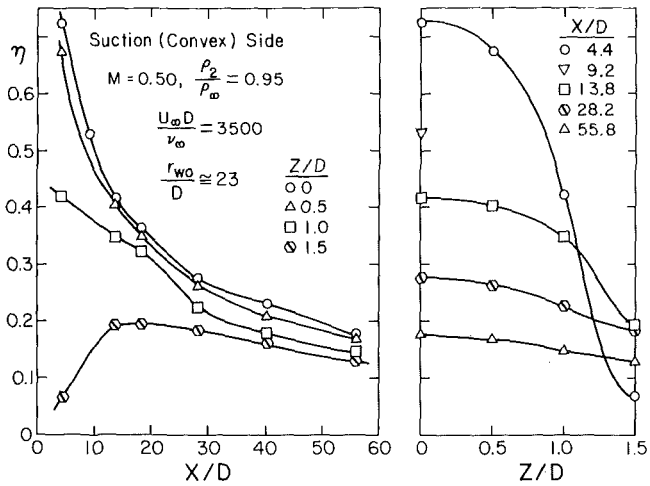

Fig. 4 Local effectiveness on convex wall, $D = 2.38$ μm, $\alpha = 35$ deg, $t = 3D$, $M = 0.5$, $\rho_2/\rho_\infty = 0.95$

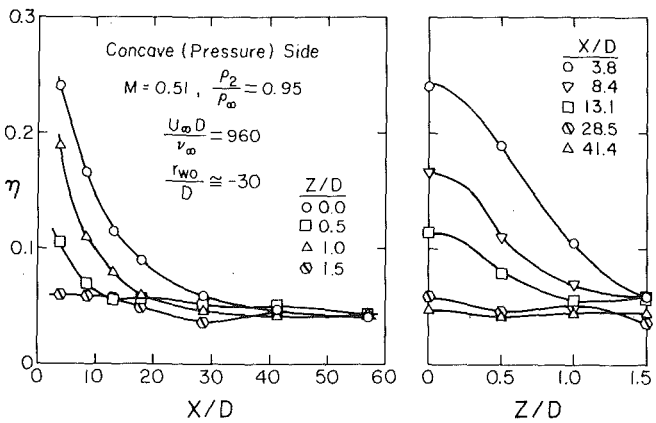

Fig. 5 Local effectiveness on concave wall, $D = 2.38$ mm, $\alpha = 35$ deg, $t = 3D$, $M = 0.51$, $\rho_2/\rho_\infty = 0.95$

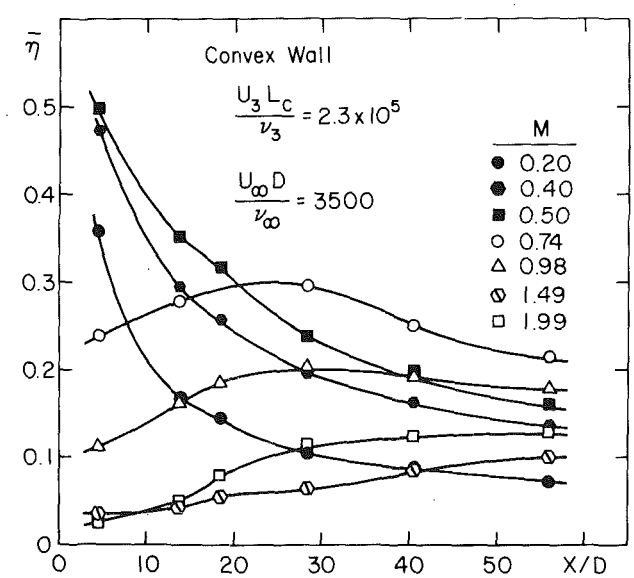

Fig. 6 Lateral average effectiveness on convex (suction) surface $\rho_2/\rho_\infty = 0.95$

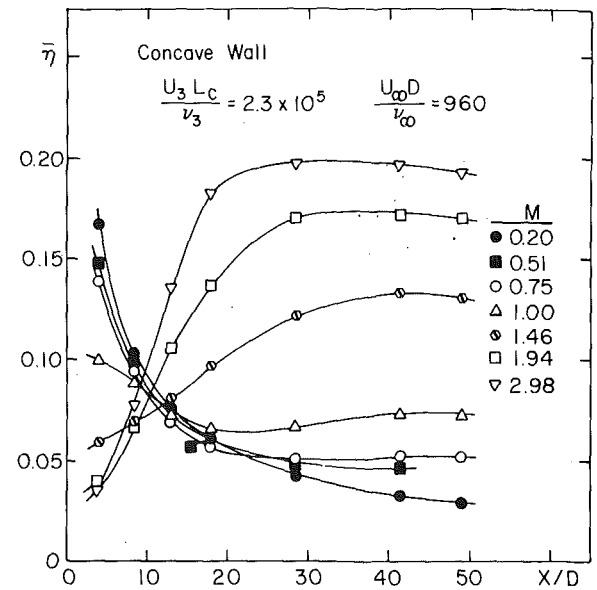

Fig. 7 Lateral average effectiveness on concave (pressure) surface, $\rho_2/\rho_\infty = 0.95$

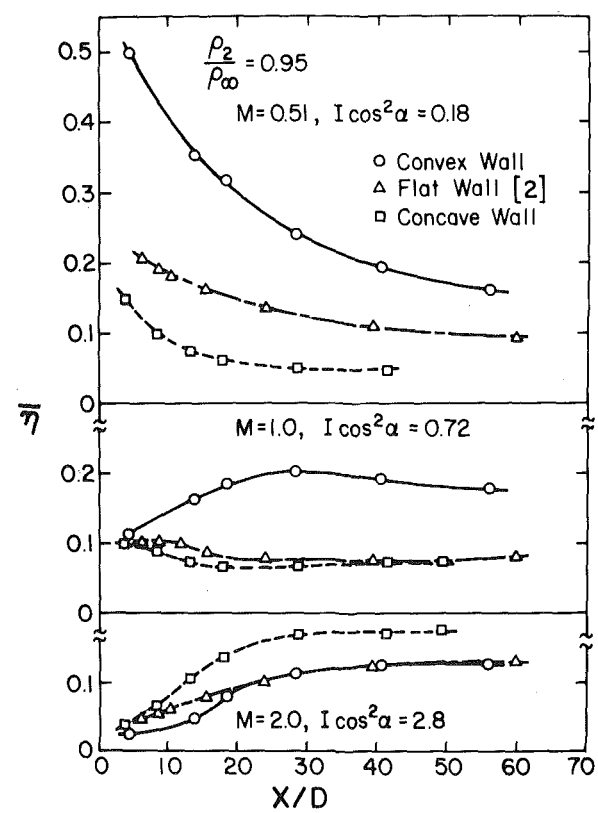

Fig. 8 Comparison of lateral average effectiveness on surfaces of different curvature, $\rho_2/\rho_\infty = 0.95$

Further downstream the effectiveness is higher at the smallest density ratio.

Conclusions

Trends in the effects of curvature on film cooling effectiveness of a jet flowing parallel to a wall can be explained by determining whether the injected fluid moves closer to or further from the wall surface. The trend of the jet trajectory can be determined by con-

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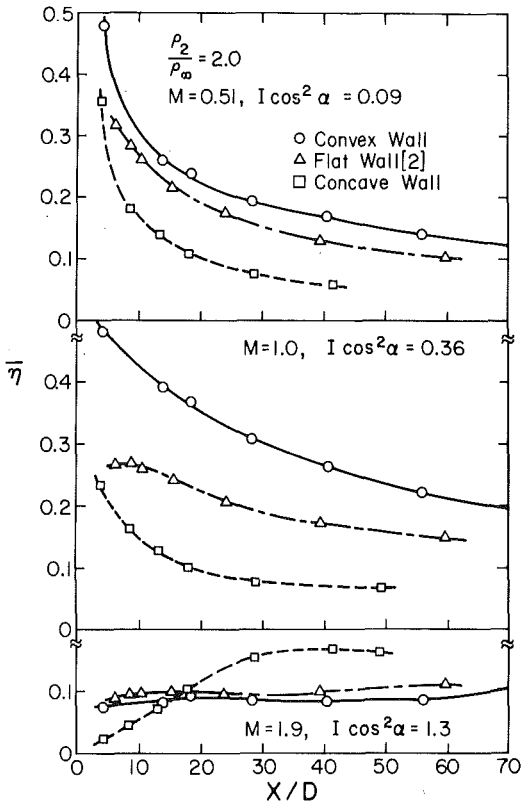

Fig. 9 Comparison of lateral average effectiveness on surfaces of different curvature, $\rho_2/\rho_\infty = 2.0$

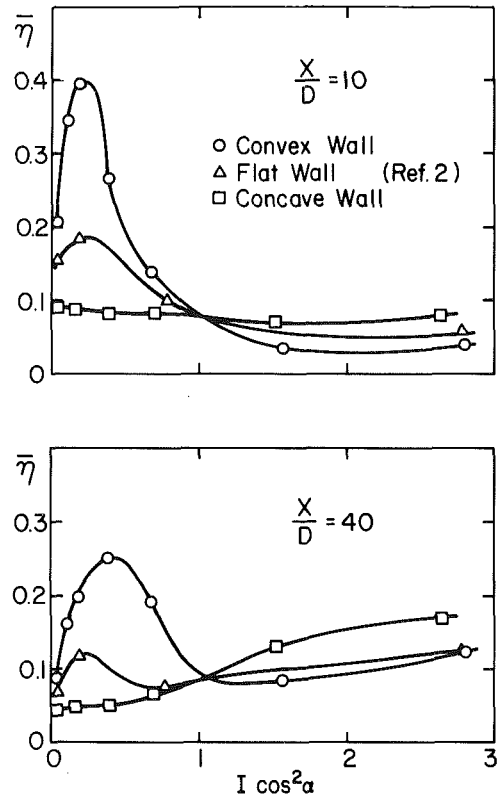

Fig. 10 Variation of lateral average effectiveness with $I \cos^2\alpha$, $\rho_2/\rho_\infty = 0.95$

sidering the balance of the forces exerted on the injected fluid of the jet by the static pressure and the centrifugal force along the path of the injected fluid. For a jet injected at an angle to a mainstream without a longitudinal pressure gradient, if $I \cos^2\alpha$ is larger than unity, the effectiveness is smaller on a convex wall, but larger on a concave wall than on a flat wall for 0 deg $\leq \alpha \leq$ 90 deg. For $I \cos^2\alpha$ less than unity the effectiveness tends to be greater on a convex surface than on a concave one.

The curvature of a surface near the injection holes or slits is particularly important to the film-cooling effectiveness. Even over a short distance downstream of the injection holes, say X/D of 5, there are substantial curvature effects on the effectiveness.

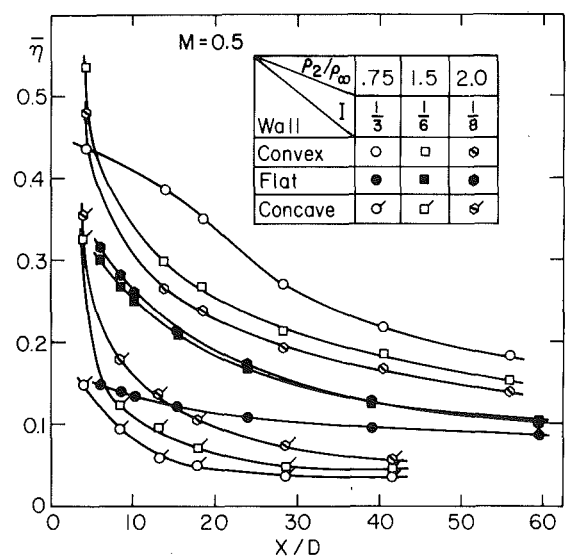

Fig. 11 Comparison of lateral average effectiveness for different density ratios on convex, flat, and concave surfaces, $M = 0.5$

Acknowledgment

The authors would like to express their appreciation to the Power Branch of the Office of Naval Research for support under contract number N00014-76-C-0246.

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